Can we detect and characterise the linguistic style of human-AI co-written text, and how does it differ from purely human or purely AI writing?

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Project Definition

Stylometry, the research focused on defining and extracting linguistic features to quantify writing style, traditionally assumes single human authorship (Stamatatos, 2009). However, recent AI detection research usually treats authorship as a binary problem, identifying human or machine-generated text. According to Lee et al. (2022), real-world writing often involves collaboration between humans and generative AI systems. Hybrid authorship, therefore, challenges both classical stylometry and current AI detection approaches. This thesis aims to investigate whether human-AI collaboration changes the overall stylistic profile of an entire text, rather than identifying individual AI-generated sentences.

The main research question is:

Can we detect and characterise the linguistic style of human-AI co-written text, and how does it differ from purely human or purely AI writing?

The sub-research questions are:

1. *Can we classify Human, AI and Hybrid as distinct categories?*
2. *How does stylometric drift vary across rewrite intensity?*
3. *Does authorship attribution remain robust under AI editing?*

Motivation

From a societal perspective, generative AI tools such as ChatGPT, Claude and Gemini are widely used for drafting, editing and rewriting text. Research done by Dwivedi (2023) shows that educational institutions, publishers and forensic linguists are subject to growing uncertainty related to authorship attribution in this hybrid context. Hence, understanding how AI intervention alters human writing style is necessary for academic integrity, authorship verification and transparency in digital communication.

From a scientific perspective, stylometry relies on the assumption that authors possess relatively stable linguistic fingerprints (Lichtblau & Stoean, 2023). If AI rewriting weakens or changes these fingerprints, this would challenge core assumptions in authorship attribution research. Hybrid texts may represent neither human nor AI style, but a structurally unique category.

Background

Recent research on AI-generated text has mainly focused on detection, rather than stylistic transformation. Many studies examine whether machine-generated text can be distinguished from human writing using statistical, stylometric and neural features. For example, Mitchell et al. (2023) proposed a method called DetectGPT, which shows that AI-generated text can often be separated from human writing

based on differences in how likely certain word sequences are. Another approach is proposed by Kirchenbauer et al. (2023), who introduce a hidden watermark that is intentionally embedded into AI-generated text, allowing it to be recognized later and enabling detection of the specific system that produced it. These studies frame the problem as detecting human and AI text in a binary setting.

However, detection robustness remains unstable. Liang et al. (2023) show that AI detection systems can be biased against non-native English writers, highlighting concerns about fairness and generalisability. Furthermore, Sadasivan et al. (2023) determine that paraphrasing and minor rewriting can significantly reduce detection accuracy. This suggests that AI intervention may weaken the distinction between human and AI writing, rather than produce easily separable categories.

While these studies address pure AI versus human distinctions, hybrid authorship introduces additional complexity. Zeng et al. (2024) investigate sentence-level detection within human-AI collaborative documents and show that hybrid texts are structurally harder to classify, particularly when human and AI contributions alternate frequently. Their findings are particularly relevant for understanding the limitations of binary detection frameworks, as hybrid writing cannot be reduced to purely human or purely AI categories.

This thesis, therefore, extends current literature by moving the focus from detection performance to stylometric drift and the development of a distinct hybrid category.

Design and Dataset Description

This thesis will include a three-class corpus consisting of human-written texts, AI-generated texts and hybrid AI-edited human texts.

Human texts will be selected from publicly available corpora, including PAN authorship datasets, TOEFL11 essays and certain verified texts from Project Gutenberg. These datasets provide variation in genre, proficiency level and author identity.

PAN authorship datasets are collections of texts, some containing over 1,000 authors, ranging from short social media posts (around 100 tokens, i.e., individual word-level units) to full-length novels exceeding 50,000 tokens. The datasets are multilingual and provide author-labeled documents with controlled training and test splits for stylometric evaluation.

The TOEFL11 corpus has 12,100 argumentative essays written by English learners from 11 native-language backgrounds. The essays are short to medium length and are controlled by a writing prompt.

The Project Gutenberg dataset contains over 60,000 full-length, single-author literary works in the public domain, such as novels, essays and short stories. These texts provide long, stylistically consistent documents.

AI texts will be generated using recent large language models such as GPT 4, Claude and Gemini under controlled prompt conditions to ensure reproducibility and alignment with the genre, length, and stylistic characteristics of the respective human datasets.

Hybrid texts will be created through manual, controlled rewriting methods. For each original human document, three levels of AI intervention will be applied to measure stylometric drift as a function of rewrite intensity:

1. AI editing (clarity, grammar and flow; minimal style change)
2. AI paraphrasing (different wording and structure; moderate style change)
3. Partial AI rewrite (rewriting sections such as the introduction or conclusion; strong style change)

For the AI-generated and hybrid texts, during generation and editing, the metadata will be stored and documented to ensure transparency, reproducibility and traceability of authorship.

Instrumentation and Material

For this research, no physical instruments are needed since the study focuses on computational analysis of textual data. The primary materials consist of the selected text corpora, including human-written texts, AI-generated texts and AI-edited versions. The analysis will be conducted using Python-based software tools for text preprocessing, feature extraction and classification.

Statistics, Algorithms and Software

All analyses will be done using Python.

Preprocessing

The same preprocessing pipeline will be used for the analysis of each text. It will consist of the following steps: lowercasing, removing extra whitespace, segmentation of sentences, tokenisation using spaCy, tagging parts of speech and lemmatization if needed for specific features.

Feature Extraction

For the purpose of this thesis, a variety of feature sets will be extracted. Character n-grams (n=3 to 5) will be used for subword stylistic patterns because they capture consistent writing patterns and are less sensitive to topic differences (Stamatatos, 2009). Function word frequencies will be calculated based on part of speech tags, including prepositions, conjunctions, pronouns, determiners and auxiliary verbs

identified using spaCy. Function words are largely content-independent and have repeatedly been shown to provide strong authorship signals (Stamatatos, 2009). The distribution of varying sentence lengths and the relative frequency of major parts of speech will also be extracted per text. The average dependency tree depth and the clause per sentence ratio will also be extracted to measure syntactic complexity using spaCy's dependency parser (Explosion, 2026).

Classification Models

This project will be treated as a supervised three-class classification problem with the labels: Human, AI-generated and Hybrid. A Logistic Regression model for character n-gram features will be used as the baseline model, as it is a standard linear classifier for high-dimensional text data (Jurafsky & Martin, 2026). A Support Vector Machine with a linear kernel will be implemented as a stylometric comparison model, since it performs well in high-dimensional feature spaces (Du et al., 2024). A Random Forest classifier will also be included to check whether non-linear feature interactions improve performance compared to linear models. All models will be trained and evaluated using the extracted stylometric features.

Evaluation

The dataset will be split into training and test sets using stratified sampling to maintain class balance. Model performance will be evaluated on a held-out test set using Accuracy, Precision, Recall and the Macro averaged F1 score to make sure that all the classes are weighted equally. Five-fold cross-validation will also be used during model development to try and prevent overfitting. Furthermore, to check if the stylistic features vary significantly between the categories, statistical analysis will be applied, but depending on how the features are distributed, the choice of method will change, therefore, it is still undecided at this stage.

Stylometric Drift Analysis

To measure stylistic change, the feature space distance between the original human texts and their AI-edited versions will be calculated using cosine similarity and Euclidean distance. This is to quantify how much stylistic shift occurs under different AI intervention levels.

Milestones and Plan

See Table 1.

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Table 1

Planned schedule for research implementation and thesis submission.

Date	Goal
2-03-2026	Decided on the research question and theoretical framework. Finalized the thesis proposal and partial literature background. Selected plausible datasets and methodological design.
9-03-2026	Finalize dataset selection criteria, collect and sample around 100 human texts, standardize formatting, organize metadata and set up the preprocessing environment.
16-03-2026	Generate AI texts under controlled prompt conditions, document prompt templates and model parameters.
23-03-2026	Create a hybrid corpus (editing, paraphrasing, partial rewriting), verify consistency and finalize the full corpus (around 500 texts).
30-03-2026	Implement the preprocessing pipeline (tokenization, POS tagging, sentence segmentation) and validate the outputs.
6-04-2026	Extract the stylometric features (character n-grams, function words, syntactic measures) and construct the feature matrices.
13-04-2026	Implement the baseline model (Logistic Regression), perform initial training and evaluation.
20-04-2026	Train the comparison models (SVM, Random Forest), perform 5-fold cross-validation and tune the parameters.
27-04-2026	Conduct stylometric drift analysis (cosine similarity, Euclidean distance) and apply statistical testing.

4-05-2026	Analyse the results, generate tables and visualizations and interpret the findings in relation to the research questions.
11-05-2026	Draft the “Results” and “Discussion” sections and refine the methodological description.
18-05-2026	Revise full thesis, finalize formatting and prepare the thesis for final submission.